

NIKHILESWARA RAO SULAKE

☎ 6281128232 | ✉ nikhil01446@gmail.com | 💻 [nikhilraosulake](https://nikhilraosulake.com) | 🌐 [Nikhil-Rao20](https://Nikhil-Rao20.github.io) | 🌐 Portfolio

Education

Rajiv Gandhi University of Knowledge and Technologies, Nuzvid IIIT

Nuzvid, Eluru

Pre University Course, CGPA-9.88

December 2021 - August 2023

Rajiv Gandhi University of Knowledge and Technologies, Nuzvid IIIT

Nuzvid, Eluru

Bachelor of Technology in Computer Science, CGPA-8.92

August 2023 - May 2027

Course Work

DRDO and DIAT, Pune - AI/ML Professional Certification Course

Online

Course Guide: Dr. Upasana Singh, Grade-9.2

June 2023 - November 2023

Experience

Neural Nurture

Remote (August 2025 - November 2025)

Research Intern - LLM

Indore, India

- Contributed to Project Lens, focusing on safeguarding against PII security breaches in large language models under both white-box and black-box settings.
- Developed and evaluated methods to identify, mitigate, and strengthen LLM privacy vulnerabilities through systematic experimentation and analysis.

Stanford Medicine Department of Radiology

Remote (June 2025 - August 2025)

Fellowship in Molecular Biology

California, USA

- Completed an intensive summer fellowship covering molecular imaging technologies, imaging agents, preclinical and clinical applications, with hands-on exposure to luminescent imaging, cellular imaging, and radiotracer development.
- Presented research on AI applications in Alzheimer's disease, highlighting contributions to brain imaging analysis during the program's final presentations.

School of Medical Science and Technology (SMST) at IIT Kharagpur

Hybrid (May 2025 - Aug 2025)

Summer Research Intern

Kolkata, India

- Developed a unified framework combining U-Net segmentation with surface topology-based reconstruction (Marching Cubes) to generate high-fidelity 3D lung models from CT scans, enhancing disease diagnosis, treatment planning, and patient-specific outcomes.
- Applied the framework to patient data, addressing CT variability and complex lung deformations, and submitted the work to *IEEE Transactions on Instrumentation and Measurement* (Impact Factor: 5.9) for publication.

Core AI

Remote (January 2025 - March 2025)

AI Researcher

Vijayawada, India

- Developed new model configurations on **Joint Autoregressive and Hierarchical Priors** for Learned **Image Compression** for image compression to work on **OCT** grayscale images.
- Developed a new methodology for efficient classification of **Alzheimer disease** on **3D Brain MRI** with visualizing the gradients using **Guided Smooth GradCam ++**.

Projects

CardioLens - Automated Segmentation of Left Ventricle and Ejection Fraction Calculation

- Developed an AI-powered echocardiogram analysis system using the Stanford's **EchoNet Dynamic** dataset to automate left ventricle segmentation and ejection fraction calculation and give out detailed health report for patients.
- Worked on Intel DPT Large and ResNet-101 models for segmentation, achieving Dice coefficients of 0.78 and 0.90, respectively and R2plus1D Model for Ejection Fraction calculation.
- Developed an algorithm to generate electrocardiogram (ECG) waveforms by analyzing heart contractions and expansions using systole and diastole boundaries.

Multi Modal Chest X-Ray Classification

- Tackled a multimodal classification challenge in **MAIC** by extracting embeddings from images (Chest X-rays) and text using the **OpenAI's CLIP model**.
- Designed a cross-attention fusion mechanism and a custom neural network, achieving a **macro-averaged F1 score of 0.35**, showcasing expertise in multimodal deep learning

Eye OCT Disease Classification and Analysis over CNN and MLP Models

- Evaluated the effectiveness of deep learning models on OCT data, achieving over 99% classification accuracy with the MLP-Mixer model.
- Leveraged techniques like confidence-uncertainty quantification to compare CNN-based models with MLP-based models for enhanced performance.

Multi backbone Integration for YOLOv8

- Developed a modular framework to integrate multiple backbones into the YOLOv8 model, replacing the **vanilla Dark-FPN** with **convolutional, Vision Mamba, transformer-based, and ViT-based** architectures.
- Implemented and analyzed these variations from scratch, leveraging model-building expertise to enhance detection performance on the Mini-COCO dataset.

Thyroid Nodule Segmentation

- Conducted extensive research on state-of-the-art models for thyroid nodule segmentation, including **UNet, UNet++, Attention UNet, Residual UNet, UNeXt, TransUNet, U²-Net, and DuckNet**.
- Evaluated and optimized segmentation models to improve dice coefficient in Thyroid Nodule Detection.

ZOYA - The AI powered Humanoid Robot

- **Project Zoya**, presented at **Teckzite 2k24**, is a **humanoid robot** capable of **voice-to-voice communication** using **GenAI**, allowing it to answer user queries with **voice-based responses**.
- Integrated the AI system with **IoT (Raspberry Pi)**, making Zoya a **self-contained, fully functional humanoid robot**, designed for interactive sessions at student events like **inaugurations and campus presentations**.

Technical Skills

- **Programming Languages:** Python, C, Java
- **Technologies:** Medical Image Analysis, Computer Vision, Deep Learning, Machine Learning, Generative AI, LLMs, VLMs
- **Frameworks:** PyTorch, Tensorflow, CUDA, OpenCV, Langchain, Hugging Face
- **Tools:** VS Code, Git, TensorBoard, Wandb, Kaggle, Draw.io
- **Environments:** Anaconda, NVIDIA Jetson Nano, NVIDIA AI Workbench, Raspberry PI
- **Soft Skills:** Communication skills, Time management, leadership

Achievements

- **SNU Multi-Modal AI 4 TB Top 22** – Led sole Indian team in chest X-ray + text analysis at Seoul National University competition 2024
- **Intel AI 2024 Hackathon – 2nd Runner-Up** – Developed AI-driven echocardiography analysis tools presented at IIT Kharagpur in IEEE Indicon 2024 2024
- **AI 4 Andhra Police Hackathon Winners** – Build a product for digitising the handwritten multi-lingual document with a user-admin website for interactions. 2025
- **NCC Innovation Challenge Finalist** – Presented women's safety project *Swetcha* to NCC leadership at Republic Day Camp, Delhi 2025
- **Parabola9 Hackathon Winner** – Built traffic analysis video captioning system with chatbot-integrated data storage 2024
- **Teckzite Expo – 2nd Place** – Created Gen-AI-powered interactive humanoid robot for student assistance 2024
- **Intel/Awiros CV Appathon Semi-Finalist** – Created hand-sign detection system for accessibility communication 2023
- **GDSC AI/ML Lead** – Directed educational sessions introducing AI concepts to 100+ students 2023–2024
- **YOLOv11 Architecture Researcher** – Published article on YOLOv11 architecture with novel diagram contribution at Analytics Vidhya Blogathon 49 (Winner) 2024